

4.2 Review

Name _____

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

Write the equation in its equivalent exponential form.

1) $\log_5 25 = 2$

1) _____

2) $\log_4 x = 3$

2) _____

3) $\log_b 49 = 2$

3) _____

Write the equation in its equivalent logarithmic form.

4) $7^2 = 49$

4) _____

5) $5^{-2} = \frac{1}{25}$

5) _____

6) $\sqrt[3]{64} = 4$

6) _____

Evaluate the expression without using a calculator.

7) $\log_4 16$

7) _____

8) $\log_{10} 10,000$

8) _____

9) $\log_4 \frac{1}{16}$

9) _____

10) $\log_5 1$

10) _____

11) $\log_5 5$

11) _____

12) $2^{\log_2 18}$

12) _____

13) $\log_3 3^{14}$

13) _____

14) $\ln \frac{1}{e^9}$

14) _____

15) $e^{\ln 136}$

15) _____

16) $\ln e^{12x}$

16) _____

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

Find the domain of the logarithmic function.

17) $f(x) = \log_5 (x + 7)$

A) $(-7, \infty)$

B) $(-\infty, 0)$ or $(0, \infty)$

C) $(7, \infty)$

D) $(5, \infty)$

17) _____

18) $f(x) = \log_7 (x - 6)$

A) $(-\infty, 0)$ or $(0, \infty)$

B) $(6, \infty)$

C) $(-\infty, 6)$ or $(6, \infty)$

D) $(-6, \infty)$

18) _____

19) $f(x) = \ln \left(\frac{1}{x - 8} \right)$

A) $(8, \infty)$

B) $(1, \infty)$

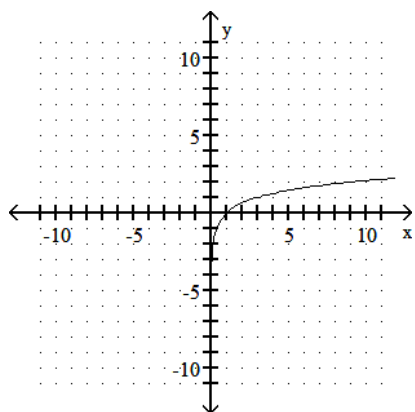
C) $(-8, \infty)$

D) $(0, \infty)$

19) _____

The graph of a logarithmic function is given. Select the function for the graph from the options.

20)



A) $f(x) = \log_3 x$

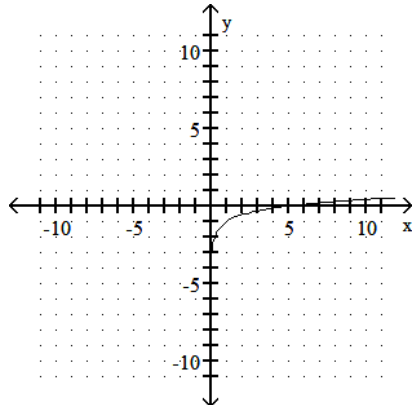
B) $f(x) = \log_3 (x + 2)$

C) $f(x) = \log_3 (x - 2)$

D) $f(x) = \log_3 x - 2$

20) _____

21)



A) $f(x) = \log_5 (x - 1)$

B) $f(x) = \log_5 x$

C) $f(x) = \log_5 x - 1$

D) $f(x) = \log_5 (x + 1)$

21) _____